Raphaël Huleux

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FIELDS

Macroeconomics, Fiscal & Monetary Policy, Inequality, Heterogeneous Agents, Computational Methods (Deep Learning)

POSITION

University of Copenhagen *2024 – Present*
Postdoctoral Researcher

EDUCATION

Copenhagen Business School *2020 – 2024*
PhD in Economics – Thesis nominated for the CBS Best PhD Thesis award

Paris School of Economics *2018 – 2020*
Master in Analysis and Policy in Economics – *With Honors*

Sciences Po Paris *2014 – 2017*
Master in Economics and Public Policy – Cum Laude, *Top 10%*

Sorbonne Paris 1 *2014 – 2017*
Master in Philosophy and Society – *With Honors*

RESEARCH PAPERS

Permanent Labor Income Inequality and Stabilization Policies (JMP)
With Eustache Elina (Banque de France)

We study how rising permanent labor income (PLI) inequality changes the transmission of monetary and fiscal policy. Standard heterogeneous-agent models imply that hand-to-mouth status is unrelated to permanent income, and that the PLI distribution is neutral for policy transmission. In the PSID, instead, the hand-to-mouth share falls from 64% to 22% across PLI quintiles. A two-asset HANK model with a non-homothetic bequest motive reproduces this gradient and breaks the neutrality. The cumulative multiplier of a fiscal package then depends more on who receives the payout and who finances it, and less on how long the deficit lasts. Applied to the One Big Beautiful Bill Act, the model gives a cumulative multiplier of -0.06*, against 0.57 in a standard HANK model with the same aggregate MPC. The effect of a rate cut depends on who receives the interest the government saves on its debt. Since 1970, rising PLI inequality has made redistributions between income groups more powerful and monetary policy more dependent on the fiscal rule.*

From Income to Wealth Inequality: Trickle-Down vs. Capital Gains
With Eustache Elina (Banque de France)
Under review, *Journal of Monetary Economics*

Higher permanent labor income inequality (PLI) raises saving, since households with higher PLI save a larger share of it. For asset markets to clear, prices must adjust: how does this feed back on wealth inequality? If the saving finances capital, interest rates fall, wages rise, and poorer households gain. If it raises asset prices instead, the capital gains accrue to the rich. In a heterogeneous-agent model with non-homothetic

wealth preferences and imperfect competition, markups weaken the first channel and let the second dominate for five decades along the transition. Fed with the observed U.S. rise in inequality since 1970, the model generates two fifths of the rise in the wealth-to-output ratio and a quarter of the rise in Tobin's Q. These price effects raise the average wealth of the top 0.1% by 25% and average wealth by 8% by 2020; under perfect competition both fall, by 4% and 5%.

Fiscal Policy in Globally Solved HANK Models with Aggregate Risk
With Jeppe Druedahl, Jacob Røpke (University of Copenhagen)

We study the role of expectations regarding aggregate risk in two standard HANK models, using a novel deep learning approach that approximates their solutions in moving-average form. In a simple Intertemporal Keynesian Cross model, we find strong sign, size and state dependence in the impulse responses to deficit-financed transfer shocks. These nonlinearities are, however, already present in the nonlinear perfect-foresight solution. Aggregate risk slightly lowers the MPC, as fewer households are hand-to-mouth. Households also stay further from the borrowing constraint, which dampens the response to negative fiscal shocks. In a larger HANK model, the nonlinearities are weaker, and the dynamics under aggregate risk and under perfect foresight again coincide.

Deep Learning Solutions of Large Non-Convex Life-Cycle Models
With Jeppe Druedahl, Jacob Røpke (University of Copenhagen)

We introduce a novel deep learning algorithm for solving large life-cycle models with both continuous and discrete choices. This allows us to simultaneously account for both labor supply choices with human capital accumulation, portfolio choices with a risky and a risk-free asset, and housing and mortgage choices. We work directly on the Bellman equation and approximate both value and policy functions with neural networks, and use a simulated training sample instead of tensor product grids. This substantially alleviates the curse of dimensionality. We demonstrate this in a consumption-saving model with multiple durable goods subject to non-convex adjustment costs where our deep learning algorithm clearly outperforms a standard value function iteration. We confirm that we can accurately solve a large life-cycle model in 3 hours on a single GPU. We solve the model simultaneously across all periods instead of with backward induction. This simplifies transfer learning, where a new solution for a new set of parameters in a calibration or estimation can easily be achieved. An accompanying easy-to-use software package implements the method.

WORK IN PROGRESS

The Exit Channel of Monetary Policy
With Eustache Elina (Banque de France), Léonard Bocquet (Amsterdam)

We study why monetary tightening can push small but very productive and fast-growing firms out of the market. These firms are highly leveraged during their growth period and are thus highly sensitive to changes in interest rates. Monetary tightening can therefore worsen the selection of firms, with long-term effects on growth and productivity.

Why is there still investment in polluting capital? Stranded assets and climate policy uncertainty

Despite governments' commitments to limit global warming to $1.5°C$*, there is still investment in carbon-intensive capital. This paper uses a growth model featuring irreversible investment, capacity utilisation, clean and polluting capital to study this apparent paradox. It shows that current investment in polluting capital and* CO_2 *emissions are coherent with expectations of a future carbon tax, if investors also expect*

a bailout of polluting capital. This result implies that governments' credibility can play an important role in reducing the cost of implementing an optimal carbon tax by committing not to bail out. However, there exists a temptation for a short-sighted government to boost output and consumption in the short run by announcing a future bailout.

Behavioral and Distributional Effects of Danish Pension Reforms (DALI)
With Jeppe Druedahl, Jacob Røpke (University of Copenhagen) and DREAM
Funded by the Economic Policy Research Network (EPRN)

Long-Run Income Expectations: A Survey Linked to Danish Registers
With Jeppe Druedahl, Manuel Menkhoff (University of Copenhagen)

VISITING POSITIONS

Banque de France, Visiting Researcher Grant, *France*	*Dec. 2025 – Jan. 2026*
IIES, *Sweden*, invited by Per Krusell	*November – December 2022*
Sciences Po Paris, *France*, invited by Xavier Ragot	*September – October 2022*

CONFERENCES and PRESENTATIONS

Danmarks Nationalbank, L-MPC workshop, *Copenhagen, Denmark*	*October 2026*
Chapel Hill–Copenhagen Conference on Macroeconomics and Deep Learning, *Chapel Hill, USA*	*September 2026*
Normac, Bergen, *Norway*	*August 2026*
CEF 2026 (Computing in Economics and Finance), *Venice, Italy*	*June 2026*
Banque de France, *Paris, France*	*January 2026*
Danmarks Nationalbank, *Copenhagen, Denmark*	*November 2025*
Normac, Copenhagen, *Denmark*	*August 2024*
Economix, *Paris, France*	*April 2024*
Macro Seminar, Copenhagen University, *Denmark*	*October 2023*
NHH, Bergen, *Norway*	*September 2023*
AFSE, Sciences Po, *Paris, France*	*June 2023*
Macroeconomics and mathematics, BI, *Oslo, Norway*	*December 2022*
Inequality and climate change workshop, CBS, *Denmark*	*November 2022*
Workshop in computational economics, CREST, *Paris, France*	*September 2022*
Macro-group at Statistics Norway, *Oslo, Norway*	*April 2022*
Danish Graduate Program in Economics (flash talk), *Denmark*	*November 2021*
Nordic Initiative for Corporate Economics (discussant), *Denmark*	*September 2021*
Danish Graduate Program in Economics (flash talk), *Denmark*	*November 2020*

TEACHING

Lecturer and coordinator, "Advanced Macroeconomics: Heterogeneous Agent Models", Graduate, *University of Copenhagen* *2026*

Lecturer, same course, with Jeppe Druedahl; rated an A course *2025*

TA, "Macroeconomics", Undergraduate, *CBS* *2022, 2024*

TA, "Macroeconomics and European Economic Integration", Undergraduate, *CBS* *2024*

TA, "Microeconomics", Undergraduate, *CBS* *2021*

Lecturer, "Mathematics for Economists", Graduate, *CBS* *2021*

TA, "Econometric Analysis of Firm Data", Graduate, *CBS* *2020, 2021*

TA, "Macroeconomics", Undergraduate, *Paris-Descartes* *2018 – 2020*

Master's thesis supervision, *CBS*: Helena Sophie Cordt (now PhD student, DTU)

Workshop on deep-learning methods for macroeconomics, *Banque de France* *2026*

GRANTS and AWARDS

Teaching bonus for an A-rated course, *University of Copenhagen* *2026*

Banque de France Visiting Researcher Grant *2025 – 2026*

Nominated, Best PhD Thesis award, *Copenhagen Business School* *2024*

LANGUAGES and SOFTWARE

French (native), English (fluent), Danish (beginner), Spanish (basic)
Python (JAX, GPU computing), R, Matlab, Dynare, Git/GitHub, LaTeX, Reuters Datastream, Bloomberg
Open-source software: EconDLSolvers, with Jeppe Druedahl and Jacob Røpke

ACADEMIC SERVICE

Referee for *Ecological Economics* *2026*

Organizer of the junior macro reading group, *KU* *2025 – 2026*

Creator and organizer of the Macro Lunches, *CBS* *2023*

Organizer (with Dolores Romero Morales) of the DGPE Workshop (100+ participants) *2023*

Main organizer of the Junior Seminar, *CBS* *2022 – 2023*

Ph.D. representative at the Ph.D. School, *CBS* *2022*

PROFESSIONAL EXPERIENCE

OFCE, Paris *September 2019 – July 2020*
Research Assistant

CFTC Workers' union, Paris *April – August 2018*
Technical advisor on economic issues

HSBC France, Paris *July – December 2017*
Intern Economist

Crédit Mutuel-CIC Market Solutions, Paris *December 2016 – July 2017*
Intern Economist

Minister of Ecology, Paris *January–July 2015*
Circular Economy Intern

REFERENCES

1. **Jeppe Druedahl**, Associate Professor, University of Copenhagen, Denmark, jeppe.druedahl@econ.ku.dk
2. **Axelle Ferrière**, Associate Professor, Sciences Po and CNRS, France, axelle.ferriere@sciencespo.fr
3. **Karl Harmenberg**, Associate Professor, IIES, Stockholm University, Sweden, karl.harmenberg@iies.su.se
4. **Pontus Rendahl**, Professor, Copenhagen Business School, Denmark, pre.eco@cbs.dk